KARL PHILIP ESPINO

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Birth Date: January 12, 2005

SUMMARY

Dedicated and hardworking student with a strong commitment to personal and professional growth. Thrives on challenges and continuously seeks opportunities to learn and improve. Highly adaptable, with a proactive mindset and a passion for tackling complex tasks. Driven by a results-oriented approach, coupled with excellent problem-solving skills and a determination to excel in any role.

EDUCATION

Mapúa University **Aug 2023 - Present**
School of Electronics, Electrical, and Computer Engineering

- Bachelor of Science and Masters in Electrical Engineering (BMEE)
  - Mapúa University School of EECE President Lister's AY 2023-2024, 1st, 2nd, and 4th Term
  - Mapúa University School of EECE President Lister's AY 2024-2025 1st, 2nd, and 3rd Term

Mapúa University **Aug 2021 - July 2023**
Senior High School Department

- Science, Technology, Engineering, and Mathematics Strand (STEM)
  - Senior High School Diploma, July 2023
    - With Highest Honors, 1st, 2nd, and 4th Quarter; With High Honors 3rd Quarter AY 2022-2023
    - With High Honors, 1st, 2nd, 3rd, and 4th Quarter, AY 2021-2022.

Justino Sevilla High School **June 2017- June 2021**

- Junior High School Diploma, July 2021
  - With High Honors, AY 2017-2018, 2018-2019, and 2020-2021; With Honors, AY 2019-2020

San Nicolas Elementary School **June 2011 - March 2017**

- Elementary Diploma, March 2017
  - School Valedictorian; Most Obedient Student; Best in Math; Best in Science; Best in Communication Art.

ACHIEVEMENTS

IIEE Excellence Awards **Present**
Award: Batch 2023 1st Rank in 1st, 2nd 3rd, and 4th Term of AY 2023-2024
Award: Batch 2023 1st Rank in 1st Term of AY 2024-2025

IIEE-Regional Math Wizard Competition - NCR **2025**
Award: 2nd Place

EECE Excellence Awards **2025**
Award: 1st Rank (Electrical Engineering Program, Batch 2023)

EECE Excellence Awards **2024**
Award: 1st Rank (Electrical Engineering Program, Batch 2023)

METROBANK-MTAP-DepEd Math Challenge **2018-2020**
Award: 3x 1st Place (Cluster 1, Elimination Round)

Math Quiz Bee Division Level - Pampanga **2017**
Award: 3rd Place

TOOLS & TECHNOLOGIES

- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint
- MATLAB
- Python
- AutoCAD
- LTSpice
- TinaPro Simulation
- CapCut
- Adobe InDesign
- Canva
- Dialux Evo

SELECTED PROJECTS

Gaming & Mental Health: Interactive Looker Studio Dashboard **March 2026**
- Built a Looker Studio dashboard correlating gaming patterns with user well-being.

Commercial Lighting Design **February 2026**
- Engineered a standard-compliant mall lighting layout using DIALux evo.

Residential Lighting Design **January 2026**
- Simulated comprehensive indoor and outdoor residential lighting in DIALux evo.

Transformer Winding Construction **July 2025**
- Designed, manually wound, and tested a functional electrical transformer.

12V DC Motor Rewinding **March 2025**
- Executed the complete rewinding and performance testing of a 12V DC motor.

PLC Traffic Control System **July 2025**
- Programmed a PLC to automate a two way-intersection traffic light sequence.

Garbage Level Monitoring System **March 2025**
- Designed a logic gate-based fill monitoring circuit featuring 7-segment visual tracking and audio alerts.

BJT Gain Optimization **November 2024**
- Calibrated BJT parameters to achieve maximum electrical signal amplification.

BJT Amplifier Circuit **October 2024**
- Designed a BJT amplifier circuit to meet specific voltage gains.

Spring Mass System Case Study **August 2024**
- Modeled and simulated dynamic mechanical responses in a spring-mass system.

Residential Home Computer Aided Drafting **May 2024**
- Drafted detailed 2D architectural and structural floor plans using AutoCAD.

EXTRACURRICULAR ACTIVITIES

Institute of Electrical and Electronics Engineers of the Philippines, Inc. - MU **Feb 2026 - Present**
- Officer, Assistant Finance Committee Head
  - Secured corporate sponsorships and managed finances for major organizational events.

The Mapúa Society of Double Degree and Joint Program **July 2025 - Present**
- Officer, Auditor
  - Audited financial records and verified receipts to ensure strict fiscal transparency.

Institute of Integrated Electrical Engineers of the Philippines, Inc. – MU **May 2024 - Present**
- Member | Former Officer, Academic Committee Head
  - Orchestrated the IIEE Excellence Awards and managed logistics to recognize top-performing students.

Mapúa University DOST Scholars' Association **Oct 2023 - Present**
- Member
  - Participated and continuously engages in DOST-related programs and activities

Physics Society of Mapua **Oct 2023 - Present**
- Member
  - Participated and continuously engages in physics-focused events.

ADDITIONAL INFORMATION

- **Technical Skills:** Strong critical thinking, problem-solving, analytical reasoning, and adaptability skills
- **Languages:** English, Filipino, Kapampangan
- **Hobbies:** *Instruments*: Piano, Guitar, and Organ; *Games*: Dota 2, CS2, ML, Valorant, Etc.
- **Activities:** Was part of a Catholic Church Choir as an Organist for several years.